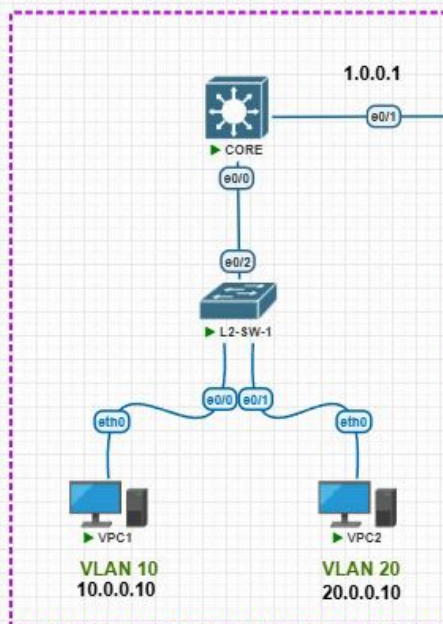
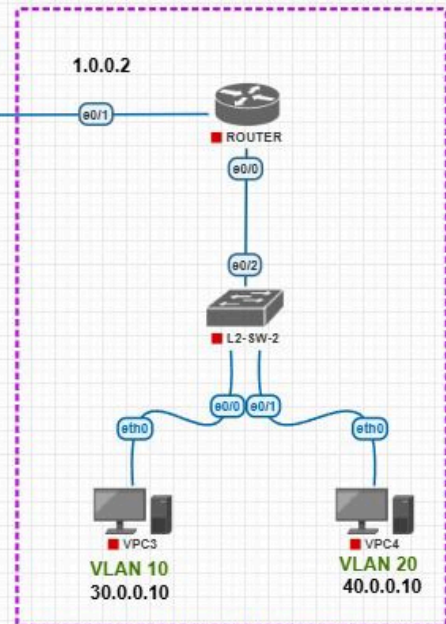




SWITCH VIRTUAL INTERFACE



ROUTER ON-A-STICK

```
L2-SW-1
!
!
interface Ethernet0/0
  switchport access vlan 10
!
interface Ethernet0/1
  switchport access vlan 20
!
interface Ethernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Ethernet0/3
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!
!
--More--
```

```
CORE
interface Ethernet0/2
!
interface Ethernet0/3
!
interface Vlan10
 ip address 10.0.0.1 255.0.0.0
!
interface Vlan20
 ip address 20.0.0.1 255.0.0.0
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end

CORE(config)#
```

```
VPC1
Invalid address

VPC1> ip 10.0.0.10/8 10.0.0.1
Checking for duplicate address...
VPC1 : 10.0.0.10 255.0.0.0 gateway 10.0.0.1

VPC1> sh run
Invalid arguments

VPC1> ping 10.0.0.1

84 bytes from 10.0.0.1 icmp_seq=1 ttl=255 time=0.654 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=255 time=1.270 ms
84 bytes from 10.0.0.1 icmp_seq=3 ttl=255 time=1.249 ms
84 bytes from 10.0.0.1 icmp_seq=4 ttl=255 time=1.378 ms
84 bytes from 10.0.0.1 icmp_seq=5 ttl=255 time=1.562 ms

VPC1> ping 20.0.0.1

84 bytes from 20.0.0.1 icmp_seq=1 ttl=255 time=2.084 ms
84 bytes from 20.0.0.1 icmp_seq=2 ttl=255 time=2.027 ms
84 bytes from 20.0.0.1 icmp_seq=3 ttl=255 time=1.591 ms
84 bytes from 20.0.0.1 icmp_seq=4 ttl=255 time=1.527 ms
84 bytes from 20.0.0.1 icmp_seq=5 ttl=255 time=1.556 ms

VPC1> ping 20.0.0.10

20.0.0.10 icmp_seq=1 timeout
84 bytes from 20.0.0.10 icmp_seq=2 ttl=63 time=3.115 ms
84 bytes from 20.0.0.10 icmp_seq=3 ttl=63 time=3.197 ms
84 bytes from 20.0.0.10 icmp_seq=4 ttl=63 time=2.772 ms
84 bytes from 20.0.0.10 icmp_seq=5 ttl=63 time=3.040 ms

VPC1> ping 20.0.0.10

84 bytes from 20.0.0.10 icmp_seq=1 ttl=63 time=2.000 ms
84 bytes from 20.0.0.10 icmp_seq=2 ttl=63 time=2.879 ms
84 bytes from 20.0.0.10 icmp_seq=3 ttl=63 time=2.437 ms
84 bytes from 20.0.0.10 icmp_seq=4 ttl=63 time=2.999 ms
84 bytes from 20.0.0.10 icmp_seq=5 ttl=63 time=1.390 ms

VPC1> █
```



```
VPC2

Press '?' to get help.

Executing the startup file

VPC2> ip 20.0.0.10/8 20.0.0.1
Checking for duplicate address...
sh runVPC2 : 20.0.0.10 255.0.0.0 gateway 20.0.0.1

VPC2> sh run
Invalid arguments

VPC2> ping 20.0.0.1

84 bytes from 20.0.0.1 icmp_seq=1 ttl=255 time=1.460 ms
84 bytes from 20.0.0.1 icmp_seq=2 ttl=255 time=1.823 ms
84 bytes from 20.0.0.1 icmp_seq=3 ttl=255 time=1.683 ms
84 bytes from 20.0.0.1 icmp_seq=4 ttl=255 time=1.352 ms
84 bytes from 20.0.0.1 icmp_seq=5 ttl=255 time=1.436 ms

VPC2> ping10.
Bad command: "ping10.". Use ? for help.

VPC2>
VPC2> ping 10.0.0.1

84 bytes from 10.0.0.1 icmp_seq=1 ttl=255 time=1.517 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=255 time=2.223 ms
84 bytes from 10.0.0.1 icmp_seq=3 ttl=255 time=2.084 ms
84 bytes from 10.0.0.1 icmp_seq=4 ttl=255 time=1.792 ms
84 bytes from 10.0.0.1 icmp_seq=5 ttl=255 time=1.489 ms

VPC2> ping 10.0.0.10

84 bytes from 10.0.0.10 icmp_seq=1 ttl=63 time=1.630 ms
84 bytes from 10.0.0.10 icmp_seq=2 ttl=63 time=2.050 ms
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=1.637 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=1.496 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=1.590 ms

VPC2> █
```

```
L2-SW-2
!
!
!
interface Ethernet0/0
  switchport access vlan 10
!
interface Ethernet0/1
  switchport access vlan 20
!
interface Ethernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Ethernet0/3
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
control-plane
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!
--More--
```

```
ROUTER
!
interface Ethernet0/0
  no ip address
  duplex auto
!
interface Ethernet0/0.1
  encapsulation dot1Q 10
  ip address 30.0.0.1 255.0.0.0
!
interface Ethernet0/0.2
  encapsulation dot1Q 20
  ip address 40.0.0.1 255.0.0.0
!
interface Ethernet0/1
  no ip address
  duplex auto
!
interface Ethernet0/2
  no ip address
  duplex auto
!
interface Ethernet0/3
  no ip address
  duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
--More--
*Aug 15 14:33:53.054: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
ipv6 ioam timestamp
!
!
!
control-plane
!
!
--More--
```

```
VPC3

VPC3> ip 30.0.0.10/8 30.0.0.1
Checking for duplicate address...
sh ipVPC3 : 30.0.0.10 255.0.0.0 gateway 30.0.0.1

VPC3> sh ip

NAME          : VPC3[1]
IP/MASK       : 30.0.0.10/8
GATEWAY       : 30.0.0.1
DNS           :
MAC           : 00:50:79:66:68:24
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPC3> ping 30.0.0.1

84 bytes from 30.0.0.1 icmp_seq=1 ttl=255 time=0.663 ms
84 bytes from 30.0.0.1 icmp_seq=2 ttl=255 time=1.109 ms
84 bytes from 30.0.0.1 icmp_seq=3 ttl=255 time=1.417 ms
84 bytes from 30.0.0.1 icmp_seq=4 ttl=255 time=1.675 ms
84 bytes from 30.0.0.1 icmp_seq=5 ttl=255 time=1.428 ms

VPC3> ping 40.0.0.1

84 bytes from 40.0.0.1 icmp_seq=1 ttl=255 time=1.868 ms
84 bytes from 40.0.0.1 icmp_seq=2 ttl=255 time=1.305 ms
84 bytes from 40.0.0.1 icmp_seq=3 ttl=255 time=0.997 ms
84 bytes from 40.0.0.1 icmp_seq=4 ttl=255 time=1.233 ms
84 bytes from 40.0.0.1 icmp_seq=5 ttl=255 time=1.212 ms

VPC3> ping 40.0.0.10

84 bytes from 40.0.0.10 icmp_seq=1 ttl=63 time=2.304 ms
84 bytes from 40.0.0.10 icmp_seq=2 ttl=63 time=2.826 ms
84 bytes from 40.0.0.10 icmp_seq=3 ttl=63 time=1.523 ms
84 bytes from 40.0.0.10 icmp_seq=4 ttl=63 time=1.551 ms
84 bytes from 40.0.0.10 icmp_seq=5 ttl=63 time=1.795 ms

VPC3> █
```



```
VPC4> ip 40.0.0.10/8 40.0.0.1
Checking for duplicate address...
ip VPC4 : 40.0.0.10 255.0.0.0 gateway 40.0.0.1

VPC4> sh ip

NAME          : VPC4[1]
IP/MASK        : 40.0.0.10/8
GATEWAY        : 40.0.0.1
DNS            :
MAC            : 00:50:79:66:68:25
LPORT          : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

VPC4> ping 40.0.0.1

84 bytes from 40.0.0.1 icmp_seq=1 ttl=255 time=1.171 ms
84 bytes from 40.0.0.1 icmp_seq=2 ttl=255 time=2.050 ms
84 bytes from 40.0.0.1 icmp_seq=3 ttl=255 time=1.706 ms
84 bytes from 40.0.0.1 icmp_seq=4 ttl=255 time=1.836 ms
ping 84 bytes from 40.0.0.1 icmp_seq=5 ttl=255 time=2.234 ms
30.0
VPC4> ping 30.0.0.1

84 bytes from 30.0.0.1 icmp_seq=1 ttl=255 time=1.804 ms
84 bytes from 30.0.0.1 icmp_seq=2 ttl=255 time=1.396 ms
84 bytes from 30.0.0.1 icmp_seq=3 ttl=255 time=1.288 ms
84 bytes from 30.0.0.1 icmp_seq=4 ttl=255 time=1.456 ms
84 bytes from 30.0.0.1 icmp_seq=5 ttl=255 time=1.167 ms

VPC4> ping 30.0.0.10

84 bytes from 30.0.0.10 icmp_seq=1 ttl=63 time=2.614 ms
84 bytes from 30.0.0.10 icmp_seq=2 ttl=63 time=1.718 ms
84 bytes from 30.0.0.10 icmp_seq=3 ttl=63 time=1.502 ms
84 bytes from 30.0.0.10 icmp_seq=4 ttl=63 time=1.624 ms
84 bytes from 30.0.0.10 icmp_seq=5 ttl=63 time=1.849 ms

VPC4> █
```

```
!
!
interface Ethernet0/0
!
interface Ethernet0/1
  no switchport
  ip address 1.0.0.1 255.0.0.0
!
interface Ethernet0/2
!
interface Ethernet0/3
!
interface Vlan10
  ip address 10.0.0.1 255.0.0.0
!
interface Vlan20
  ip address 20.0.0.1 255.0.0.0
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
```

```
ROUTER
!
interface Ethernet0/0
  no ip address
  duplex auto
!
interface Ethernet0/0.1
  encapsulation dot1Q 10
  ip address 30.0.0.1 255.0.0.0
!
interface Ethernet0/0.2
  encapsulation dot1Q 20
  ip address 40.0.0.1 255.0.0.0
!
interface Ethernet0/1
  ip address 1.0.0.2 255.0.0.0
  duplex auto
!
interface Ethernet0/2
  no ip address
  duplex auto
!
interface Ethernet0/3
  no ip address
  duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
```

```
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1>
VPC1> ping 20.0.0.10

84 bytes from 20.0.0.10 icmp_seq=1 ttl=63 time=2.285 ms
84 bytes from 20.0.0.10 icmp_seq=2 ttl=63 time=2.921 ms
84 bytes from 20.0.0.10 icmp_seq=3 ttl=63 time=2.199 ms
84 bytes from 20.0.0.10 icmp_seq=4 ttl=63 time=2.242 ms
84 bytes from 20.0.0.10 icmp_seq=5 ttl=63 time=1.863 ms

VPC1> ping 30.0.0.10

84 bytes from 30.0.0.10 icmp_seq=1 ttl=62 time=5.871 ms
84 bytes from 30.0.0.10 icmp_seq=2 ttl=62 time=3.742 ms
84 bytes from 30.0.0.10 icmp_seq=3 ttl=62 time=3.120 ms
84 bytes from 30.0.0.10 icmp_seq=4 ttl=62 time=3.283 ms
84 bytes from 30.0.0.10 icmp_seq=5 ttl=62 time=3.443 ms

VPC1> ping 40.0.0.10

84 bytes from 40.0.0.10 icmp_seq=1 ttl=62 time=5.399 ms
84 bytes from 40.0.0.10 icmp_seq=2 ttl=62 time=3.313 ms
84 bytes from 40.0.0.10 icmp_seq=3 ttl=62 time=2.997 ms
84 bytes from 40.0.0.10 icmp_seq=4 ttl=62 time=2.569 ms
84 bytes from 40.0.0.10 icmp_seq=5 ttl=62 time=2.894 ms

VPC1> █
```